

Metaverse—you cannot escape the term—and surprisingly, you cannot define it either because it is evolving so rapidly, bringing more and more technologies and use-cases under its wing each day! Some call the metaverse a parallel universe, but it is not, because it is interleaved with the real one. Some call the metaverse a make-believe world, but in fact, it is not, because real people are doing real business and investing real money in it. So, let us just accept the fact that there is no single holistic definition for metaverse. It is an evolution that we have to understand component by component, and piece together the way we want it to work for us.

Although the term metaverse was coined three decades ago in a sci-fi novel, it gained recognition only recently, with the surge in popularity of massively multiplayer online (MMO) role-playing games like *Second Life*, *Fortnite*, *Sandbox*, *Enjin*, *Decentraland*, *Minecraft*, and *Axie Infinity*.

MMO games have evolved to a point where the boundaries between their virtual worlds and the real world are blurring. In fact, it is unfair to call them games anymore, because they have transformed into interoperable virtual platforms where millions of people interact with each other, play games or just chill with friends, own and sell properties, create and trade things, and even perform or attend concerts!

Realising the untapped potential of this nascent marketplace, real-world brands like *Taco Bell*, *Pizza Hut*, and *Adidas* have started marketing their goods in virtual worlds, and even opening showrooms there. Companies, hospitals, universities, and other organisations have also created virtual places to connect with their stakeholders, for various purposes like training, collaboration, or simply bonding! With covid physically separating people, they are turning to the metaverse to bring them together.

So, from being a place for geeks to hangout, the metaverse has now become an advanced ‘fusion world’ that brings together the best of real and virtual worlds.

Once again, what’s metaverse?

Imagine the internet in three dimensions (3D)—you being able to move inside it, like you would in your physical world. That is, probably, where the metaverse is taking us. It makes use of advanced telecom infrastructure, artificial intelligence (AI), and extended reality (XR)—that is, virtual reality (VR), augmented reality (AR), mixed reality (MR), and the likes—to create online platforms that are more immersive, interactive, and interoperable than they are today.

These online platforms or virtual worlds can be accessed through apps on your mobile phone, computer, or other digital devices. You will be represented by your *avatar* in the metaverse. With VR gear, you can immerse yourself completely in the virtual world, but considering its prohibitive cost, many platforms just let users engage through touchscreens. The metaverse also has dynamic digital marketplaces, driven by the forces of demand and supply.

If you want to draw a quick, very basic parallel with the real world, money translates to digital currencies in the metaverse, while assets translate into non-fungible tokens (NFTs). Blockchain technology forms the backbone of these marketplaces, with the tokens and transactions being recorded in it. Real money can be converted to digital currencies through exchanges, and real assets can also be recorded as NFTs in the blockchain. Here is where the blurring of boundaries really happens!

We have been covering all these aspects of the metaverse in detail in the past few issues of *Electronics For You*. Here, we walk through a couple of exciting and impactful use-cases of the metaverse.

Telemedicine: Try it on the twin first

The metaverse can bridge global divides in the availability of reliable medical care. In many remote regions, there are no skilled medical practitioners and people have to travel several kilometres to get proper medical care. Telemedicine helps not only in such situations where care is not available locally, but also in cases where a complex condition requires an expert’s intervention—and the expert is somewhere across the world.

The situation has improved remarkably post-Covid, with several medical practitioners getting used to treating simple conditions through phone and video calls. According to a *Forbes* report, just 43% of healthcare facilities had the ability to provide remote treatment to patients before 2020, but today that figure stands at 95%. In the coming years, the metaverse will add quality to quantity, making these online consultations more immersive and interactive. Virtual reality gives a feeling of being in a real clinic, improving the connect between a patient and a doctor. Agreed, VR gear is not yet within reach of the masses, but a small setup in each village is enough to cater to all the families there. Like internet cafes took the Web to the masses, perhaps virtual clinics will take expert treatment to them.

The metaverse can also help with psychological counselling and caregiving. Take the case of a recuperating patient, who requires daily physiotherapy sessions but is unable to travel to a healthcare facility. The therapist can extend the treatment to the patient remotely, using a metaverse space, VR gear, and haptic feedback suits. One space where doctors are betting big on the metaverse is in helping patients with phobias. In the metaverse, doctors can expose patients to situations and things they fear or are allergic to, under controlled environments. They can monitor how